

CS 499 Module One Assignment Aiden Villanueva

Complete this template by replacing the bracketed text with the relevant information.

Project Proposal - GaiaML: A Machine Learning Framework for Exoplanet Candidate Detection Using Gaia DR3 Public Data

I. Self-Introduction: Address all of the following questions to introduce yourself.

- A. How long have you been in the Computer Science program?

I am in my last 4 terms at SNHU in the BSCS program, so around 3 years.

- B. What have you learned while in the program? List three of the most important concepts or skills you have learned.

While I have gone from barely knowing any coding to being able to create my own games in multiple languages, I would not say that learning coding languages has been the most important part of the last three years. Overall, the most impactful things I learned were how to research, how to debug and how to plan. Learning how to research and teach myself has opened so many doors that I would have avoided before. I now have full confidence when faced with a new technology, technique or challenge that I will be able to seek out the information I need and distill it down to my application.

Likewise, knowing how to find bugs, decode heavy error reports, research solutions and lookup documentation is the most efficient way to handle bugs. Instead of poking around the problem, you can narrow down causes and follow a logical process to isolate issues and confront them one at a time.

Before starting this program, one of my biggest struggles was starting projects and becoming bogged down in details or allowing scope creep to consume them. Now, I start all of my projects by creating enough planning documentation to prevent stagnation and meandering. If I have a question, the answer is likely noted nearby. If I feel confused about the logic flow in my program, I can pull up the flowchart and see at a glance where I am and where I'm headed.

- C. Discuss the specific skills you aim to demonstrate through your enhancements to reach each of the course outcomes.

Through these planned enhancements, I intend to demonstrate software engineering, algorithm design, machine learning, database management, and technical communication skills. I will redesign the original CS-370 reinforcement learning project into a *modular machine learning application* capable of training, validating, and evaluating models using astronomical data, specifically from the Gaia mission's data release 3 (DR3). The project will demonstrate professional software architecture, efficient data processing, algorithm selection, model evaluation, documentation, and secure handling of scientific datasets. These enhancements will also showcase my ability to communicate complex technical concepts through diagrams, code reviews, and written analysis while making appropriate design tradeoffs.

- D. How do the specific skills you will demonstrate align with your career plans related to your degree?

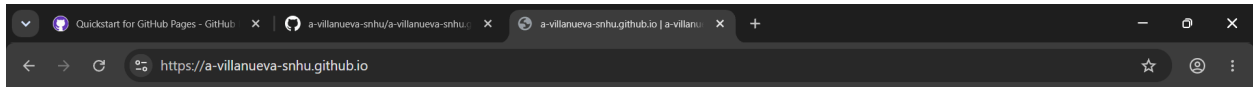
My long-term goal is to pursue graduate study in artificial intelligence while working in software engineering or scientific computing, particularly with the vast amount of new astronomical data from telescopes like Gaia, James Webb and Vera C. Rubin. The skills demonstrated through this enhancement project, including machine learning implementation, software architecture, data engineering, and model evaluation, closely match the work performed in AI research and data-driven software development. Completing this project will strengthen my portfolio by demonstrating that I can design, build, and document a complete machine learning application rather than simply implementing isolated algorithms.

- E. How does this contribute to the specialization you are targeting for your career?

This project supports my goal of specializing in artificial intelligence and scientific computing. Rather than focusing solely on web or business software, the project applies machine learning techniques to a real scientific dataset, demonstrating the ability to solve research-oriented problems using modern AI methods. It also provides experience with data pipelines, feature engineering, model evaluation, and software design practices commonly used in AI development.

II. ePortfolio Set Up:

- A. Submit a **screen capture** of your ePortfolio GitHub Pages home page that clearly shows your URL.
- You already have a repository in GitHub where you uploaded projects in previous courses. Your ePortfolio will reside in GitHub but can link to work at other sites, such as Bitbucket.
- B. Use the GitHub Pages link in the Resource section for directions on:
- How to create your GitHub website and publish code to GitHub Pages
 - Issues, such as adding links to other sites
- C. Paste a screenshot of your GitHub Pages home page with your URL clearly showing in the space below.



a-villanueva-snhu

a-villanueva-snhu.github.io

My ePortfolio

III. Enhancement Plan:

A. **Category One:** Software Engineering and Design

- i. **Select an artifact** that is **aligned with the** software engineering and design **category** and explain its origin. Submit a file containing the code for the artifact you choose with your enhancement plan.

The artifact comes from my CS-370 Artificial Intelligence course, where I implemented the reinforcement learning logic for a pirate treasure hunting agent using instructor-provided scaffolding. For the capstone, the project will be transformed into a standalone machine learning application for identifying potential exoplanet host stars using publicly available Gaia DR3 astronomical data.

Note: Your artifact may be work from the following courses:

- IT 145: Foundation in Application Development
- CS 250: Software Development Lifecycle
- CS 260: Data Structures and Algorithms
- IT 315: Object Oriented Analysis and Design
- CS 320: Software Testing, Automation, and Quality Assurance
- CS 330: Computational Graphics and Visualization
- CS 340: Advanced Programming Concepts
- CS 350: Emerging Systems Architectures and Technologies
- CS 360: Mobile Architecture and Programming
- IT 365: Operating Environments
- IT 380: Cybersecurity and Information Assurance
- CS 405: Secure Coding

- CS 410: Reverse Software engineering
- IT 340: Network and Telecommunication Management
- IT 380: Cybersecurity and Information Assurance

- ii. **Describe** a practical, well-illustrated **plan** for enhancement in alignment with the category, including a pseudocode or flowchart that illustrates the planned enhancement.

The first phase of the enhancement will focus on transforming the original Jupyter Notebook into a maintainable software application. I will begin by separating the existing notebook into reusable Python modules responsible for configuration, data loading, preprocessing, model training, evaluation, and utility functions. After establishing the project structure, I will implement a command-line interface that allows users to configure training parameters without modifying source code. Logging and exception handling will then be added to improve debugging and usability, followed by automated testing of the major components. Finally, I will document the application and organize the repository into a professional GitHub project suitable for continued development.

The enhancements for category one will follow this logical flow:

Create Project Structure -> Separate Notebook into Modules -> Implement CLI -> Implement Logging -> Implement Testing -> Generate Documentation -> Release Initial Version

For this category of enhancement, consider improving a piece of software, transferring a project into a different language, reverse engineering a piece of software for a different operating system, or expanding a project's complexity. These are just recommendations. Consider being creative and proposing an alternative enhancement to your instructor.

Think about what additions to include to complete the enhancement criteria in this category. Since one example option is to port to a new language, that is the kind of scale that is expected. This does not mean you need to port to a new language but instead have an equivalent scale of enhancement. Underlying expectations of any enhancement include fixing errors, debugging, and cleaning up comments, but these are not enhancements themselves.

- iii. Explain how the planned enhancement will **demonstrate** specific **skills** and align with course outcomes.
 - a. Identify and describe the specific skills you will demonstrate that align with the course outcome.

Specific skills that will be demonstrated in this section are:

Object-oriented programming to keep the project clean and repeatable. Software architecture through designing systems with intent and under project constraints. Maintainability, Testing and Documentation. Version-controlled repository organized according to common software engineering practices, including documentation, modular source organization, testing, and reproducible project configuration.

- b. Select one or more of the course outcomes below that your enhancement will align with.

- **Design and evaluate computing solutions.**
- **Demonstrate innovative techniques and tools.**
- **Professional written and visual communication.**

Expected outcome: Upon completion of this enhancement, the project will be easier to maintain, easier to extend with future machine learning models, and easier for other developers to understand, reproduce, validate and extend. These improvements transform the original classroom assignment into a reusable software application that better reflects professional software engineering practices.

Course Outcomes:

1. Employ strategies for building collaborative environments that enable diverse audiences to support organizational decision-making in the field of computer science.
2. Design, develop, and deliver professional-quality oral, written, and visual communications that are coherent, technically sound, and appropriately adapted to specific audiences and contexts.
3. Design and evaluate computing solutions that solve a given problem using algorithmic principles and computer science practices and standards appropriate to its solution while managing the trade-offs involved in design choices.
4. Demonstrate an ability to use well-founded and innovative techniques, skills, and tools in computing practices for the purpose of implementing computer solutions that deliver value and accomplish industry-specific goals.
5. Develop a security mindset that anticipates adversarial exploits in software architecture and designs to expose potential vulnerabilities, mitigate design flaws, and ensure privacy and enhanced security of data and resources.

B. Category Two: Algorithms and Data Structures

- i. **Select an artifact** that is **aligned with the** algorithms and data structures **category** and explain its origin. Submit a file containing the code for the artifact you choose with your enhancement plan. You may choose work from the courses listed under Category One.

Artifact: Same Artifact.

- ii. **Describe** a practical, well-illustrated **plan** for enhancement in alignment with the category, including a pseudocode or flowchart that illustrates the planned enhancement.

The second enhancement phase focuses on replacing the original reinforcement learning implementation with a supervised machine learning pipeline. I will first evaluate the Gaia DR3 dataset to determine which stellar properties are appropriate predictive features. After cleaning missing

values and removing unusable observations, I will engineer additional features where appropriate and divide the dataset into training and testing subsets. An XGBoost classifier will then be trained and evaluated using multiple performance metrics. Finally, the model will be tuned through hyperparameter adjustment and validated against known exoplanet host stars to measure prediction quality.

The core pipeline plan will be as follows:

**Acquire Gaia Dataset -> Validate Records -> Clean Missing Values ->
Feature Engineering -> Split Training/Test Data -> Train XGBoost ->
Evaluate Model -> Tune Parameters -> Generate Predictions**

For this category of enhancement, consider improving the efficiency of a project or expanding the complexity of the use of data structures and algorithms for your artifact. These are just recommendations. Consider being creative and proposing an alternative enhancement to your instructor. Note: You only need to choose one type of enhancement per category.

Think about what additions to include to complete the enhancement criteria in this category. Since one example option is to port to a new language, that is the kind of scale that is expected. Perhaps you might increase the efficiency and time complexity of an algorithm in an application and detail the logic of the increased time complexity. Remember, you do not need to port to a new language but instead have an equivalent scale of enhancement. Underlying expectations of any enhancement include fixing errors, debugging, and cleaning up comments, but these are not enhancements themselves.

- iii. Explain how the planned enhancement will **demonstrate** specific **skills** and align with course outcomes.
 - a. Identify and describe the specific skills you will demonstrate to align with the course outcome.

Skills demonstrated:

**Machine learning algorithms
Complexity analysis
Efficient data structures
Performance evaluation**

- b. Select one or more of the course outcomes listed under Category One that your enhancement will align with.

Design and evaluate computing solutions.

Innovative computing techniques.

Expected outcome: Upon completion of this enhancement, the original reinforcement learning assignment will have evolved into a supervised machine learning pipeline capable of processing astronomical observations, training an XGBoost classification model, and generating candidate prediction scores along with evaluation metrics that support model validation and future refinement. The software will be capable of ingesting scientific data, analyzing it through XGBoost gradient boosted trees and exporting transparency data and final candidate prediction scores.

C. Category Three: Databases

- i. **Select an artifact** that is **aligned with the** databases **category** and explain its origin. Submit a file containing the code for the artifact you choose with your enhancement plan. You may choose work from the courses listed under Category One.

Artifact: Same Artifact

The database enhancement will focus on designing a reusable data management layer for the Gaia dataset. Rather than reading raw CSV files during every execution, the application will import selected Gaia observations into a relational (MySQL) database for efficient storage and retrieval. A preprocessing stage will validate records, remove incomplete observations, and normalize the data before it is stored. Queries will then retrieve filtered datasets for machine learning while allowing processed results to be exported for future experiments. This enhancement demonstrates database design, data validation, query optimization, and data pipeline development.

Specific Enhancements:

MySQL database
CSV import/export
Missing value handling
Query CRUD CLI
Pipeline automation

- ii. **Describe** a practical, well-illustrated **plan** for enhancement in alignment with the category, including a pseudocode or flowchart that illustrates the planned enhancement.

This section will be designed with the following flow in mind:

**Download Gaia Data -> Import into Database ->
Validate Records -> Clean Dataset -> Normalize Fields ->
Store Processed Data ->
Query Training Set -> Export Dataset ->
Machine Learning Pipeline**

For this category of enhancement, consider adding more advanced concepts of MySQL, incorporating data mining, creating a MongoDB interface with HTML/JavaScript, or building a full stack with a different programming language for your artifact. These are just recommendations; consider being creative and proposing an alternative enhancement to your instructor. Note: You only need to choose one type of enhancement per category.

Think about what additions to include to complete the enhancement criteria in this category. Since one example option is to port to a new language, that is the kind of scale that is expected. Perhaps you might increase the efficiency and time complexity of an algorithm in an application and detail the logic of the increased time complexity. Remember, you do not need to port to a new language but instead have an equivalent scale of enhancement. Underlying expectations of any enhancement include fixing errors, debugging, and cleaning up comments, but these are not enhancements themselves.

- iii. Explain how the planned enhancement will **demonstrate** specific **skills** and align with course outcomes.
 - a. Identify and describe the specific skills you will demonstrate that align with the course outcome.

Skills demonstrated by this category include: Data engineering and Database interaction, working with Scientific datasets and Data validation

- b. Select one or more of the course outcomes listed under Category One that your enhancement will align with.

Demonstrate an ability to use well-founded and innovative techniques, skills, and tools in computing practices for the purpose of implementing computer solutions that deliver value and accomplish industry-specific goals. Demonstrate an ability to use well-founded and innovative techniques, skills, and tools in computing practices for the purpose of implementing computer solutions that deliver value and accomplish industry-specific goals.

Employ strategies for building collaborative environments that enable diverse audiences.

Expected outcome: Upon completion of this enhancement, GaiaML will provide a reusable workflow for importing, validating, storing, and preparing Gaia DR3 observations for machine learning experiments. The completed system will automate repetitive preprocessing tasks while providing a consistent and reproducible dataset for model training and evaluation. The application will retrieve Gaia DR3 observations through either automated queries using astroquery or locally downloaded datasets, depending on dataset size and project requirements. Lastly, GaiaML will store all inputs, weight histories, feature metadata and outcomes for future experimentation.

IV. ePortfolio Overall Skill Set

- A. Accurately describe the **skill set** to be illustrated by the **ePortfolio overall**.
 - i. Skills and outcomes planned to be illustrated in the code review

The code review will demonstrate my ability to analyze existing software, identify limitations, justify architectural improvements, and explain how the project evolved from a classroom reinforcement learning exercise into a modular machine learning framework. It will highlight software engineering practices, algorithm selection, testing strategies, and design tradeoffs.

- ii. Skills and outcomes planned to be illustrated in the narratives

The narratives will document each enhancement phase, describing the engineering decisions, implementation process, challenges encountered, and lessons learned. They will explain why specific technologies and algorithms were selected and how each enhancement improves functionality, maintainability, and performance while aligning with the course outcomes.

- iii. Skills and outcomes planned to be illustrated in the professional self-assessment

My professional self-assessment will reflect on the growth I experienced throughout the Computer Science program, emphasizing my transition from learning programming fundamentals to designing complete software systems. The assessment will highlight my strengths in software engineering, artificial intelligence, scientific computing, and technical communication while discussing how this project supports my long-term goal of pursuing graduate study and a career in AI research and scientific software development.